From Long Term Classroom Programs and Medium / Short Classroom Program 4 in Top 10, 10 in Top 20, 43 in Top 100, 75 in Top 200, 159 in Top 500 Ranks & 3542 total selections in IIT-JEE 2012

ALL INDIA TEST SERIES

FIITJEE

JEE (Advanced), 2013

FULL TEST – II

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 237**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into three sections: **Section-A, Section-B & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 – 06)** contains 6 multiple choice questions which have only one correct answer. Each question carries **+5 marks** for correct answer and **– 2 marks** for wrong answer.
Section-A (07 – 12) contains 2 paragraphs. Based upon paragraph, 3 multiple choice questions have to be answered. Each question has only one correct answer and carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

(ii) **Section-B (01 – 02)** contains 2 Matrix Match Type questions containing statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. Each question carries **+8 marks** for all correct answer. For each correct row **+2 marks** will be awarded. There may be one or more than one correct choice. No marks will be given for any wrong match in any question. There is no negative marking.

(iii) **Section-C (01 – 05)** contains 5 Numerical based questions with single digit integer as answer, ranging from 0 to 9 and each question carries **+3 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A
Single Correct Choice Type

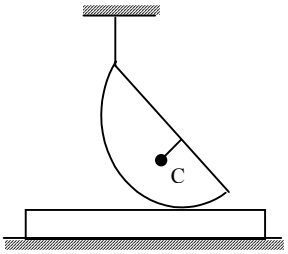
This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. In a series LCR circuit, if V is the root mean square (rms) value of the applied voltage. Across resistor (R) inductor (L) and capacitor (C) the rms values of the potential difference are V_R, V_L and V_C, respectively.

(A) $V = V_R + V_L + V_C$ (B) $V^2 = V_R^2 + V_L^2 + V_C^2$

(C) $V^2 = V_R^2 + (V_L - V_C)^2$ (D) $V^2 = V_L^2 + (V_R - V_C)^2$

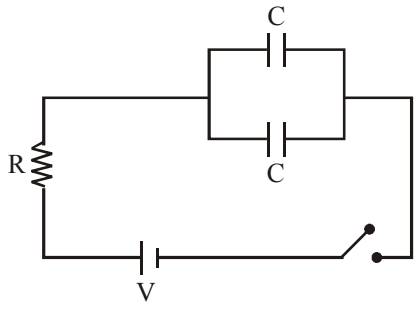
2. Lower surface of a plank is rough and lies over a rough horizontal surface. Upper surface of the plank is smooth and has a smooth hemisphere placed over it through a light string as shown. After the string is burnt trajectory of C.M. of sphere is



(A) circle (B) ellipse

(C) straight line (D) none of these

3. If switch S closed at t = 0 then work done by battery upto time t is equal to



(A) $\frac{CV^2}{2}\left(1-e^{-\frac{2t}{CR}}\right)$

(B) $CV^2\left(1-e^{-\frac{t}{2CR}}\right)$

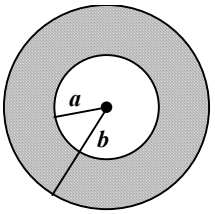
(C) $CV^2\left(1-e^{-\frac{t}{CR}}\right)$

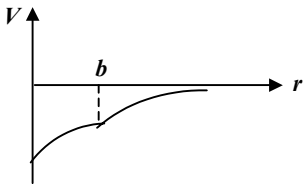
(D) None

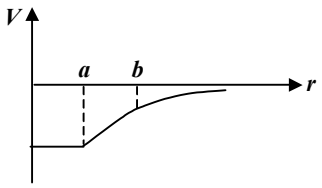
Rough work

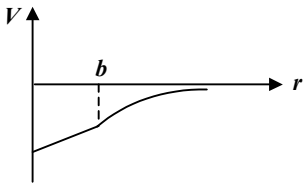
4. Circular scale of a screw guage moves through 4 divisions of main scale in one rotation. If the number of divisions on the circular scale is 200 and each division of the main scale is 1 mm., the least count of the screw guage is
(A) 0.01 mm (B) 0.02 mm
(C) 0.03 mm (D) 0.04 mm

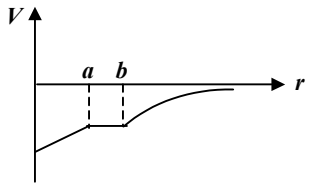
5. A sphere of mass *M* and radius *b* has a concentric cavity of radius *a* as shown in figure. The graph showing variation of gravitational potential *V* with distance *r* from the center of sphere is



(A)


(B)


(C)


(D)


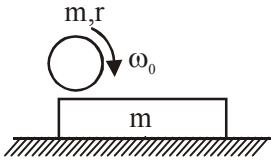
6. Let F_{pp}, F_{pn} and F_{nn} denote the magnitudes of the net force by a proton on a proton, by a proton on a neutron and by a neutron on a neutron respectively. Neglect gravitational force. When the separation is 1 fm,
(A) $F_{pp} > F_{pn} = F_{nn}$ (B) $F_{pp} = F_{pn} = F_{nn}$
(C) $F_{pp} > F_{pn} > F_{nn}$ (D) $F_{pp} < F_{pn} = F_{nn}$

Rough work

Comprehension Type

This section contains 2 groups of questions. Each group has 3 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

Paragraph for Question Nos. 7 to 9



In the figure shown a plank of mass m is lying at rest on a smooth horizontal surface. A disc of same mass m and radius r is rotated to an angular speed ω_0 and then gently placed on the plank. If we consider the plank and the disc as a system then frictional force between them is an internal force. Momentum of the system changes due to external force only. It is found that finally slipping cease. Assume that plank is long enough. μ is coefficient of friction between disc and plank.

7. Final velocity of the plank is

(A) $\frac{r\omega_0}{4}$ (B) $\frac{r\omega_0}{\sqrt{10}}$

(C) $\frac{r\omega_0}{2}$ (D) $\frac{r\omega_0}{2\sqrt{10}}$

8. Time when slipping ceases

(A) $\frac{r\omega_0}{2\mu g}$ (B) $\frac{r\omega_0}{\sqrt{10}\mu g}$

(C) $\frac{r\omega_0}{4\mu g}$ (D) $\frac{r\omega_0}{2\sqrt{10}\mu g}$

9. Magnitude of the change in angular momentum of the disc about center of mass of the disc

(A) $\frac{3}{4}mr^2\omega_0$ (B) $\frac{1}{4}mr^2\omega_0$

(C) zero (D) $\frac{1}{2}mr^2\omega_0$

Rough work

Paragraph for Question Nos. 10 to 12

As a charged particle q moving with a velocity $\vec{v}$ enters a uniform magnetic field $\vec{B}$, it experiences a force $\vec{f} = q(\vec{v} \times \vec{B})$. For θ = 0º or 180º, θ being the angle between $\vec{v}$ and $\vec{B}$, force experienced is zero and the particle passes undeflected. For θ = 90º, the particle moves along a circular arc and the magnetic force (qvB) provides the necessary centripetal force $\left(\frac{mv^2}{r}\right)$. For other values of θ ($\theta \neq$ 0º, 180º, 90º), the charged particle moves along a helical path which is the resultant motion of simultaneous circular and translational motions.

Suppose a particle, that carries a charge of magnitude q and has a mass 4×10^{-15} kg, is moving in a region containing a uniform magnetic field $\vec{B} = -0.4\hat{k}$T. At a certain instant, velocity of the particle is $\vec{v} = (8\hat{i} - 6\hat{j} + 4\hat{k}) \times 10^6$ m/s and force acting on it has a magnitude 1.6 N.

Answer the following:

10. Which of the three components of acceleration have non–zero values ?
    (A) x and y (B) y and z
    (C) z and x (D) x, y and z

11. Which of the following is correct?
    (A) Motion of the particle is non–periodic but y and z – position co–ordinates vary in a periodic manner
    (B) Motion of the particle is non–periodic but x and y – position co–ordinates vary in a periodic manner
    (C) Motion of the particle is non–periodic but x and z – position co–ordinates vary in a periodic manner
    (D) Motion of the particle is periodic and all the position co–ordinates vary in a periodic manner

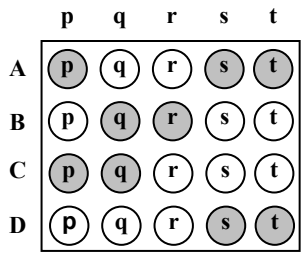
12. If the co–ordinates of the particle at t = 0 are (2m, 1m, 0), co–ordinates at a time t = 3T, where T is the time period of circular component of motion, will be
    (A) (2m, 1m, 400m) (B) (0.142m, 120m, 0)
    (C) (2m, 1m, 1.884m) (D) (142m, 130m, 628m)

Rough work

SECTION - B

Matrix – Match Type

*This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:*
If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:



1. Match the following.

Column A		Column B	
(A)	If the work done by force in cyclic path is zero, the force is	(p)	Non conservative
(B)	If the work done by a force in cyclic path is not zero, the force is	(q)	Negative
(C)	Work done by friction force can be	(r)	Conservative
(D)	Work done by spring force can be	(s)	Positive

2. Match the following.

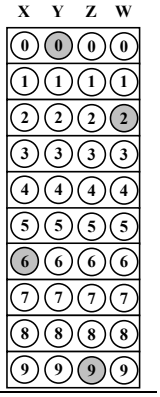
Column A		Column B	
(A)	Amplitude of constituent wave	(p)	0.06
(B)	Position of node at x = … m	(q)	0.5
(C)	Position of antinode at x = … m	(r)	0.25
(D)	Amplitude at $x = \left(\frac{3}{4}\right)$ m	(s)	0.03

Rough work

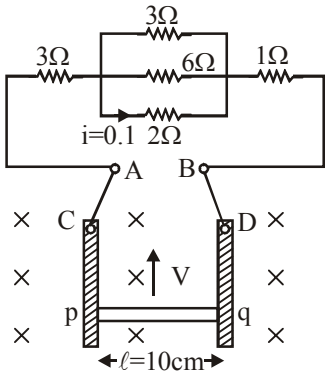
SECTION – C

Integer Answer Type

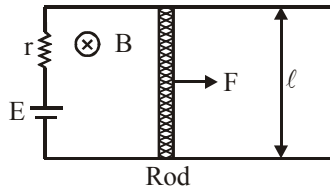
This section contains 5 questions. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.



1. A conducting rod Pq is sliding on two parallel conducting rods with constant velocity in a uniform magnetic field of induction B = 2 Wb/m^2 as shown in the figure. The rods are connected with a circuit. What should be velocity of rod Pq (in m/s) if current in 2Ω resistor is 0.1 A ?

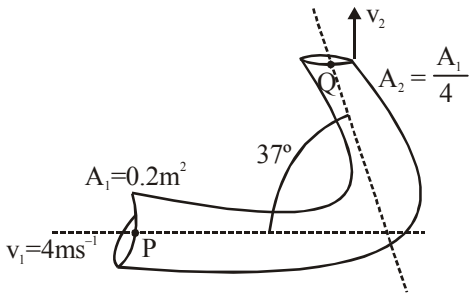


2. Two parallel conducting rails are connected to a source of emf E and internal resistance r. Another conducting rod of length ℓ having negligible resistance lies at rest and can slide without friction over the rails. A uniform magnetic field B is applied perpendicular to the plane of the rails. At t = 0, the rod is pulled along the rails by applying a force F. The velocity of the rod is observed to be $v = v_0 \cos(\omega t)$ then find the power (in watt) spent by the force over 1 cycle.
(Given B = 2 Tesla, $r = 2 \times 10^{-4}\Omega$, $v_0 = 2\ ms^{-1}$, ℓ = 1cm)

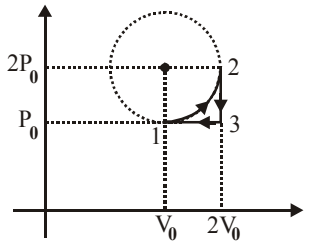


Rough work

3. Oil enters the bend of a pipe in the horizontal plane with velocity 4 ms^{-1} and pressure 280 × 10^3 Nm^{-2} as shown in the figure. The pressure of oil at the point Q is 86λ. Find the value of λ. (KNm^{-2}).
(Take specific gravity of oil as 0.9 and sin 37° = 0.6)



4. One mole of ideal monoatomic gas is taken along a cyclic process as shown in the figure. Process 1 → 2 shown is 1/4th part of a circle as shown by dotted line process 2 → 3 is isochoric while 3 → 1 is isobaric. If efficiency of the cycle is n% where n is an integer. Find n.



5. A spring of force constant k = 300 N/m connects two blocks having masses 2 kg and 3 kg, lying on a smooth horizontal plane. If the spring block system is released from a stretched position, the number of complete oscillations in 1 minute is 6n. Find the value of n. Take $\pi = \sqrt{10}$.

Rough work

Chemistry PART – II

SECTION – A

Straight Objective Type

This section contains 6 multiple choice questions numbered 1 to 6. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. Which test can distinguish 1,1 dichloroethane and 1,2 dichloroethane.

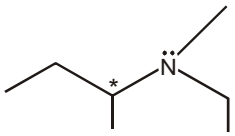
   (A) (i) aq KOH, (ii) 2,4 - DNP (B) $NaHSO_3$

   (C) Na_2CO_3 (D) $FeCl_3$ solution

2. Calculate the potential of an indicator electrode versus the standard hydrogen electrode, which originally contained 0.1 M MnO_4^- and 0.1 M H^+ and which was treated with Fe^{+2} necessary to reduce 90% of $KMnO_4$ to Mn^{+2}. $E^0_{MnO_4^-/Mn^{+2}} = 1.51$ v

   (A) 1.3 (B) 1.43

   (C) 1.48 (D) 1.4

3. 

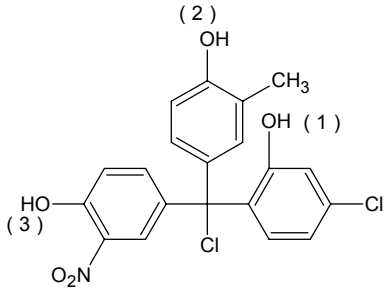
   Let the starred carbon in the given amine has 'S' configuration. What is the isomeric relationship between the two forms of this compound that are inter converted by amine inversion.

   (A) Diastereomers (B) Enantiomers

   (C) Identical (D) Functional isomers

Rough Work

4.

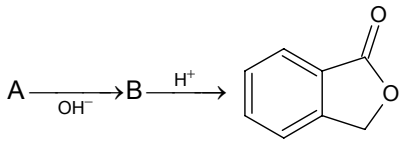


Acidity of H is in the order

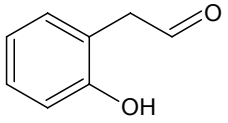
(A) $2 > 1 > 3$ (B) $3 > 1 > 2$

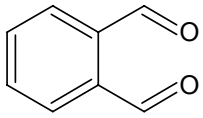
(C) $1 > 2 > 3$ (D) $2 > 3 > 1$

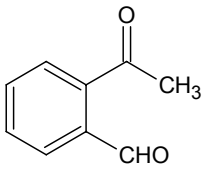
5.



Possible structure for compound A is

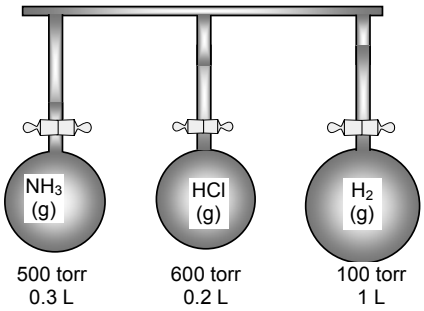
(A) 

(B) 

(C) 

(D) None of these

6.



What is the total pressure after all the stopcocks are opened at room temperature? (considering the volume of connecting tubes negligible

(A) 246.66 torr (B) 86.66 torr

(C) 100 torr (D) 1200 torr

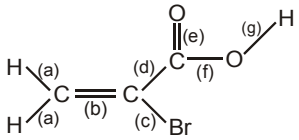
Rough Work

PARAGRAPH TYPE

This section contains 2 groups of questions. Each group has 3 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

Paragraph for Question Nos. 7 to 9

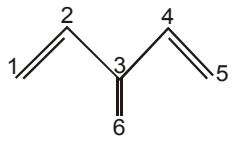
Bond length is an important criterion in deciding the reactivity of organic entities. Bond length is affected by a number of factors like atomic size electronegativity, bond order and resonance.

7. 

For the given resonating structure of the molecule find the correct trend of bond length.

(A) e < b < g < a < f < d < c (B) a < g < b < e < d < f < c

(C) g < a < e < b < f < d < c (D) b < e < d < f < g < a < c

8. 

Taking into account all the resonating structures of the above molecule, find which bond has highest bond order :

(A) $C_1 – C_2$ (B) $C_2 – C_3$

(C) $C_3 – C_6$ (D) (a) and (c) are having same bond order.

9. How many different Mn-O bond length(s) is/are there in $KMnO_4$ molecule.

(A) 1 (B) 2

(C) 3 (D) 4

Rough Work

Paragraph for Question Nos. 10 to 12

The solubility of an ionic compound depends upon lattice energy and hydration energy of that compound. Lattice energy involves the fact that how tightly are ions held together in ionic lattice whereas hydration energy involves the attraction of ions towards water molecule to get themselves hydrated. The dissolution occurs when hydrator energy predominates over lattice energy. The solubility also depends upon this difference. More is $\left(\Delta H_h - \Delta H_i\right)$ more is solubility.

10. Select the correct choice :
    (A) $KHCO_3$ is less soluble in water than $NaHCO_3$.
    (B) CaC_2O_4 is soluble in water and alkalies.
    (C) $Ca(HCO_3)_2$ an obtained in solid state.
    (D) BeF_2 is water soluble but CaF_2 is water insoluble.

11. Select the incorrect choice :
    (A) solubility of alkaline earth metal's carbonates, sulphates and chromates decreases from Be to Ba.
    (B) solubility of alkaline earth metal's hydroxides is less than alkali metal hydroxides.
    (C) solubility of alkaline earth metal's oxides increases from Be to Ba.
    (D) SO_2 on passing in lime water turns is milky.

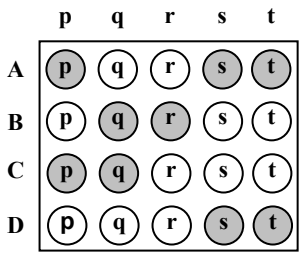
12. Report the correct order for lattice energy :
    (A) BeO > MgO > CaO
    (B) $BeF_2 < MgF_2 < CaF_2$
    (C) AgF < AgCl < AgBr
    (D) $Fe_2O_3 < FeO$

Rough Work

SECTION-B
(Matrix Type)

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:



1. Match the following

	Column – I		Column – II
(A)	$Be(OH)_2 < Mg(OH)_2 < Ca(OH)_2 < Sr(OH)_2 < Ba(OH)_2$	(p)	Solubility in water
(B)	$BeCO_3 > MgCO_3 > CaCO_3 > SrCO_3 > BaCO_3$	(q)	Thermal stability
(C)	$Li(OH) < NaOH < KOH < RbOH < CsOH$	(r)	Hydration energy
(D)	$Li_2CO_3 < Na_2CO_3 < K_2CO_3 < Rb_2CO_3 < Cs_2CO_3$	(s)	Lattice Energy
		(t)	Basic strength in aqueous solution

2. Match the following

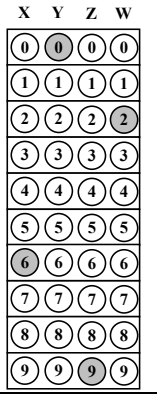
	Column – I (Using Pt electrode)		Column – II
(A)	Aqueous solution of HCl	(p)	O_2 evolved at anode
(B)	Aqueous solution of NaCl	(q)	H_2 evolved at cathode
(C)	Molten NaCl	(r)	Cl_2 evolved at anode
(D)	Aqueous solution of $AgNO_3$	(s)	Na deposited at cathode
		(t)	Ag deposited at cathode

Rough Work

SECTION – C

Integer Answer Type

This section contains 5 questions. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.



1. When CH_3CHO is treated with drops of concentration H_2SO_4 a pleasant smelting liquid paraldehyde results. The number of aldehyde molecules needed to from one molecule of paraldehyde is

2. If number of planes of symmetry in hexagonal unit cell is 'n' then the value of 'n' is:

3. If aqueous solution of NaCl is electrolyzed completely at 25°C, the total number of gases evolved is

4. 500 ml of 0.1 M $CaBr_2$(density of $CaBr_2$ solution = 1.02 gm/ml) is mixed with 500 ml of 0.2 M $AgNO_3$ (density of $AgNO_3$ solution = 1.034 gm/ml) solution. The mass of water in the mixture is 'x' gm. Then $\frac{x}{200}$ is

5. How many optically active alkanes are possible with molecular formula C_7H_{16}?

Rough Work

Mathematics PART – III

SECTION – A

Straight Objective Type

This section contains 6 multiple choice questions numbered 1 to 6. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. Let $g(x) = f_n(x)$, where $f_2(x) = f(f(x))$, $f_3(x) = f(f(f(x)))$ ..... and $f(x) = x^3 - \frac{3}{2}x^2 + x + \frac{1}{4}$ then $\int_{1/4}^{3/4} g(x)\,dx$ is

(A) $\frac{n}{2} + 2$ (B) $2^n + 1$

(C) $\frac{1}{4}$ (D) $\frac{3}{4}$

2. The area of the region bounded between the curves $y = e\left\|x\right|\ln\left|x\right\|$, $x^2 + y^2 - 2\left(\left|x\right| + \left|y\right|\right) + 1 \geq 0$ and x-axis where $|x| \leq 1$, if α is the x-coordinate of the point of intersection of curves in 1st quadrant, is

(A) $4\left[\int_0^{\alpha} ex\ln x\,dx + \int_{\alpha}^{1}\left(1 - \sqrt{1-(x-1)^2}\right)dx\right]$ (B) $4\left[\int_0^{\alpha} ex\ln x\,dx - \int_{1}^{\alpha}\left(1 - \sqrt{1-(x-1)^2}\right)dx\right]$

(C) $2\left[-\int_0^{\alpha} ex\ln x + \int_{\alpha}^{1}\left(1 - \sqrt{1-(x-1)^2}\right)dx\right]$ (D) $2\left[\int_0^{\alpha} ex\ln x + \int_{\alpha}^{1}\left(1 - \sqrt{1-(x-1)^2}\right)dx\right]$

3. The value of $\lim_{n\to\infty}\sum_{r=1}^{n}\left(\sum_{t=o}^{r-1}\frac{1}{5^n}\cdot {}^nC_r \cdot {}^rC_t \cdot 3^t\right)$ is equal to

(A) 4 (B) 3

(C) 1 (D) none of these

4. 8 spheres of radius 1 unit is kept on a table with their centres at the vertices of a regular octagon and each sphere touching its two neighbours. If a sphere is placed in the centre on the table touching all of the 8 sphers, then its radius is

(A) $\frac{1}{\sqrt{2}}$ (B) $1 - \frac{1}{\sqrt{2}}$

(C) $\sqrt{2}$ (D) $1 + \frac{1}{\sqrt{2}}$

Rough work

5. If f(x) = $(2x-3\pi)^{25}+\frac{4}{3}x+\cos x$ and g(x) is the inverse of f(x), then find $\frac{d}{dx}(g(x))$ at x = 2π

(A) $\frac{7}{3}$ (B) $\frac{3}{7}$

(C) $\frac{30\pi^{25}+4}{3}$ (D) none of these

6. Three friends whose ages form a G.P. divide a certain sum of money in proportion to their ages. If they do that three years later, when the youngest is half the age of the oldest, then he will receive Rs. 150/– more than he gets now, and the middle friend will get Rs. 15 more than now, then the age of the youngest friend is

(A) 27 (B) 18

(C) 14 (D) 12

Comprehension Type

This section contains 2 groups of questions. Each group has 3 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

Paragraph for Question Nos. 7 to 9

Read the following write up carefully and answer the following questions:

Let $<a_n>$ and $<b_n>$ be the arithmetic sequences each with common difference 2 such that $a_1 < b_1$ and let $c_n = \sum_{k=1}^{n} a_k$, $d_n = \sum_{k=1}^{n} b_k$. Suppose that the points $A_n(a_n, c_n)$, $B_n(b_n, d_n)$ are all lies on the parabola C : $y = px^2 + qx + r$ where p, q, r are the constants.

7. The value of p equals

(A) $\frac{1}{4}$ (B) $\frac{1}{3}$

(C) $\frac{1}{2}$ (D) 2

8. The value of q equals

(A) $\frac{1}{4}$ (B) $\frac{1}{3}$

(C) $\frac{1}{2}$ (D) 2

Rough work

9. If r = 0 then the value of a_1 and b_1 are

(A) $\frac{1}{2}$ and 1 (B) 1 and $\frac{3}{2}$

(C) 0 and 2 (D) $\frac{1}{2}$ and 2

Paragraph for Question Nos. 10 to 12

Read the following write up carefully and answer the following questions:
Let y = f(x) be a curve such that $x = 1 - 3t^2$, $y = t - 3t^3$, where t is a parameter and f(x) is not constantly zero $\forall\ t \in R$. The tangent drawn at P(t) to the curve makes an angle θ with positive direction of x-axis.

10. If the tangent at $A \equiv (-2, 2)$ meet the curve again at B then coordinates of point B is

(A) $\left(-3, -\frac{9}{2}\right)$ (B) $\left(-\frac{1}{3}, -\frac{2}{9}\right)$

(C) $\left(-\frac{1}{3}, \frac{2}{9}\right)$ (D) $\left(\frac{1}{3}, -\frac{2}{9}\right)$

11. The acute angle between the tangents at x = 0 to the curve y = f(x) is

(A) 0 (B) $\frac{\pi}{6}$

(C) $\frac{\pi}{4}$ (D) $\frac{\pi}{3}$

12. Let h(x) = |f(x)| + f(x) and g(x) = |f(x)| – f(x), then $\int_a^b h(x)dx$ is equal to

(A) $\int_b^a g(x)dx$ (B) $\int_a^b g(x)dx$

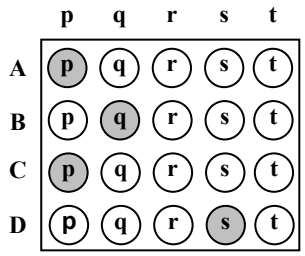
(C) $\int_b^a g(x)dx$ only if $a, b \in [0, 1]$ (D) none of these

Rough work

SECTION – B

(Matrix Type)

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE** statement in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:
If the correct matches are A – p; B – q; C – p; and D – s; then the correct darkening of bubbles will look like the following:



1. Match the following column–I with column–II.

	Column – I		Column – II
(A)	The equation of the right bisector plane of the segment joining (2, 3, 4) and (6, 7, 8) is	(p)	$\frac{13}{\sqrt{21}}$
(B)	The equation of the plane through the point (1, 2, –3) which is parallel to the plane $3x – 5y + 2z = 11$ is given by	(q)	$\frac{7}{\sqrt{21}}$
(C)	The distance of the point (2, 1, –1) from the plane $x – 2y + 4z = 9$ is	(r)	$\frac{20}{\sqrt{21}}$
(D)	The line $\frac{x-4}{1}=\frac{y-2}{1}=\frac{z-k}{2}$ lies completely on the plane $2x – 4y + z = 7$, then the value of $\frac{k}{\sqrt{21}}$ is	(s)	$x + y + z – 15 = 0$
		(t)	$3x – 5y + 2z + 13 = 0$

2. Match the following column–I with column–II.

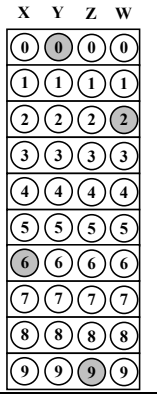
	Column – I		Column – II
(A)	The number of integers in the domain of $f(x)=\frac{1}{\sqrt{\ln\cos^{-1}x}}$, is	(p)	0
(B)	If the angle between the plane $x – 3y + 2z = 1$ and the line $\frac{x-1}{2}=\frac{y-1}{1}=\frac{z-1}{-3}$ is θ, then the value of cosec θ is	(q)	1
(C)	Let $J=\int_{-5}^{-4}(3-x^2)\tan(3-x^2)dx$ and $K=\int_{-2}^{-1}(6-6x+x^2)\tan(6x-x^2-6)dx$, then (J + K) equals	(r)	2
(D)	The value of a for the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ (a > b), if the extremities of the latus–rectum of the ellipse having positive ordinate lies on the parabola $x^2=-2(y-2)$, is	(s)	4
		(t)	6

Rough work

SECTION – C

Integer Answer Type

This section contains 5 questions. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.



1. If the line $\frac{x-1}{2} = \frac{y+2}{3} = \frac{z-4}{6}$ intersect the xy, yz and zx planes at A, B and C respectively, and if volume of the tetrahedron OABD is V, where 'O' is origin and D is the image of C in the x–axis, then the value of [V] is _____. (Where [.] denote greatest integer function).

2. Let $1+\sum_{r=1}^{10}\left(3^r \cdot {}^{10}C_r + r \cdot {}^{10}C_r\right) = 2^{10}(\alpha \cdot 4^5 + \beta)$ where $\alpha, \beta \in N$ and $f(x) = x^2 - 2x - k^2 + 1$.If α, β lies between the roots of $f(x) = 0$, then find the smallest positive integral value of k.

3. Let x be the 7th term from the beginning and y be the 7th term from the end in the expression of $\left(3^{1/3} + \frac{1}{4^{1/3}}\right)$. If $\frac{x}{y} = \frac{1}{12}$ then the value of n is _____.

4. Let z_1, z_2, z_3 be complex numbers of unit modulus such that $|z_1 - z_2|^2 + |z_1 - z_3|^2 = 4$ then $|z_2 + z_3|$ is equal to _____.

5. A student is allowed to select at most 'n' books from a collection of 2n + 1 books. If the total number of ways in which a student selects atleast one book is 63 then the value of n is _____.

Rough work